

Nocturnal Visual Place Recognition for Campus-Scale Robot Localization

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Background

We study **day-to-night visual place recognition (VPR)** for robot localization in GPS-denied environments. The task is to determine whether a live camera image corresponds to a place already stored in a daytime reference map. This is difficult because nighttime scenes change appearance substantially and small viewpoint changes can alter the ranking of visually similar places.



Figure 1: Example day reference image and night query from the campus dataset.

Methodology

Our evaluation combines a standard benchmark flow with a campus-specific relabeling study. The key observation is that the original campus protocol assumed **one night image should match one exact day image**. In practice, several daytime images can be valid views of the same place.

We therefore study a two-phase pipeline:

1. **Offline map building:** collect day images or a traversal video, sample reference frames, extract descriptors, and save a reusable map.
2. **Online localization:** extract a descriptor from each live frame, retrieve the top- k closest references, and apply a thresholded match decision.

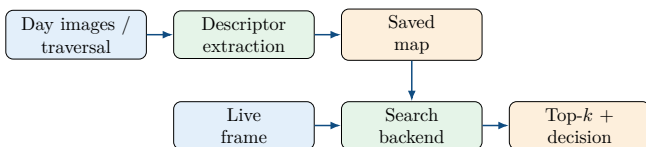
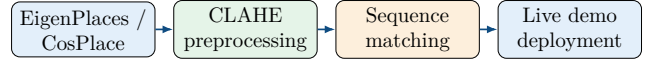


Figure 2: High-level architecture of the live VPR system.

Our Solution

We use a **training-free pipeline** built around strong pre-trained descriptors.



- **Baseline:** strong global descriptors.
- **CLAHE:** reduce the day-night appearance gap.
- **Sequence:** improve ranking with temporal context.
- **Live demo:** run from webcam, phone, or robot stream.

GardensPoint suggests a **ranking problem**: the right place is often in the top-5, but not always first.

For campus evaluation, we treat relabeling as an **evaluation fix**, not the main contribution. After manual review, night queries were matched to **place IDs** rather than one exact day image so same-place, different-view retrievals are credited fairly.

Results

The original campus setup was not robust to multiple valid views under a strict image-level evaluation protocol.

The earlier benchmark was under-crediting correct same-place retrievals. After relabeling the data at the **place level**, the strongest descriptors improved substantially:

Model	AUC	R@1	Note
CosPlace	0.627	0.969	strongest top-1
EigenPlaces	0.663	0.922	best AUC
PatchNetVLAD	0.641	0.938	strong alternative
HDC-DELF	0.507	0.922	cleaner thresholding

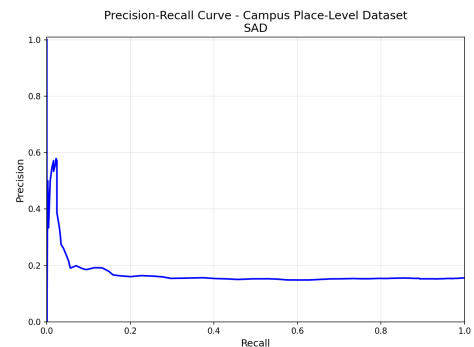


Figure 3: Place-level PR curve summary on the campus dataset.

The relabeling changed the story in an important way:



Figure 4: Qualitative same-place retrieval examples under place-level evaluation.

- **CosPlace**: R@1 improved from **0.727** to **0.969**
- **NetVLAD**: R@1 improved from **0.394** to **0.891**
- **EigenPlaces**: AUC improved from **0.443** to **0.663**

This indicates that many earlier “false matches” were not truly wrong places. They were often **same-place, different-view matches** that the strict labeling protocol did not credit.

Future Research

- Add **sequence-aware localization** so consecutive frames reinforce one another.
- Incorporate **motion and pose prediction**, for example with a Kalman-filter-style state estimate.
- Add a dedicated **open-set benchmark split** with truly unmatched night queries after place-level review.
- Strengthen **live robot deployment** using the TurboPi camera stream.

References

Berton et al., *CosPlace*, CVPR 2022. Berton et al., *EigenPlaces*, ICCV 2023. Arandjelović et al., *NetVLAD*, CVPR 2016. Milford and Wyeth, *SeqSLAM*, ICRA 2012.